Mohammad Khaki Bakhtiarvand

Computer Vision Engineer · Python Backend Engineer

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PROFESSIONAL SUMMARY

Software engineer focused on **Computer Vision**, **Deep Learning** and real-time video analytics. Built detection pipelines with **YOLO** and **OpenCV**, concurrent multi-camera **RTSP** ingestion, parallel processing architectures and centralized **GPU** inference. Professional backend experience with **Python**, **Django** and **FastAPI**, backed by solid software architecture, database transaction, authentication and asynchronous processing skills. Aims to build intelligent products on computer vision and deep learning.

COMPUTER VISION PROJECTS

Automatic License Plate Recognition (ALPR) System

Deep Learning · YOLO · OpenCV

Python | YOLO | OpenCV | RTSP | Multiprocessing | Multithreading | Shared Memory | Redis

- Designed and developed a **deep-learning-based** plate recognition system covering plate localization, character detection and **final plate string reconstruction**.

Multi-camera & GPU architecture

- Built a parallel multi-camera architecture with **Multiprocessing** and **Multithreading** to ingest and process **RTSP** streams concurrently; **configuration-driven** via a **JSON** file for camera count and processing settings, with no code changes.
- Implemented **centralized GPU inference**: the **YOLO** model is loaded once and serves inference requests from multiple cameras.
- Developed inter-process frame transfer with **Shared Memory** and a lock-free **Seqlock** buffer, reducing communication overhead and avoiding processing of stale frames.
- Designed and evaluated **benchmark** scenarios comparing multiprocessing and multithreading layouts, from **1×30** to **6×5** (processes × threads).

Detection accuracy & stability

- Developed a **Plate Stabilizer** using **IoU tracking** and **majority voting** to stabilize results across consecutive frames, with an independent ring buffer per vehicle.
- Built a dedicated **character detector** with **max_det=300** to stop characters being dropped in dense scenes.
- Fixed plate-validation logic that wrongly discarded plates containing certain Persian letters, including **"Alef" (الف)**.

Reliability & testing

- Designed a **Detection → Verification → Stabilization → Deduplication → Alert** pipeline.
- Used **Redis** with a **1-hour TTL** to suppress duplicate alerts, and a **ThreadPoolExecutor** for non-blocking API calls.
- Wrote **unit tests** that validate components without physical cameras or a GPU.

VizGuard / Wn.vizguard · Industrial Computer Vision

YOLO · OpenCV · RTSP

- Developed industrial computer-vision solutions for **visual monitoring**, vehicle traffic management and safety-event detection.
- **VizGuard:** license plate recognition and traffic management with plate detection and **blacklist** matching.
- **Wn.vizguard: fire and smoke detection** with a **custom YOLO model**, generating alerts with detection confidence.
- Processed surveillance camera streams and turned model output into events usable by monitoring systems.

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Skills & Professional Experience

TECHNICAL SKILLS

Computer Vision & Deep Learning
YOLO · OpenCV · License Plate Recognition (ALPR) · Object Detection · Character Detection · IoU Tracking · GPU Inference · RTSP Streaming · Multi-camera Processing

Python & Software Engineering
Python · OOP · SOLID · Type Hints · Asyncio · Multiprocessing · Multithreading · Exception Handling · Logging

Backend Development
Django · Django REST Framework · FastAPI · REST API Design · JWT Authentication · Clean Architecture · Hexagonal Architecture

Database & Distributed Processing
PostgreSQL · SQLAlchemy · Redis · RabbitMQ · Celery · Database Transactions · Query Optimization · Connection Pooling

Infrastructure & Testing
Linux · Docker · Nginx · Git · GitHub Actions · Pytest · Unit Testing · Integration Testing

Machine Learning
Scikit-learn · KNN · Regression · Feature Engineering · Data Preprocessing · Model Evaluation

PROFESSIONAL EXPERIENCE

Backend Engineer (Django / DRF) · Tagavi Print House, Iran — 2024 – 2025

- Designed and built the **customer and admin panel APIs** of a **CRM** platform for the printing industry with **Django** and **DRF**, including business workflows.
- Redesigned the accounting and payment system around **data consistency** and **idempotency**, using **database transactions**, **transaction isolation levels** and **SELECT FOR UPDATE** to prevent **race conditions** and **partial writes**.
- Optimized database performance with **select_related** and **prefetch_related**, eliminated **N+1 queries** and managed connections with **PgBouncer**.
- Implemented **Redis-based idempotency** and **rate limiting** to block duplicate request processing and control per-user request rates.
- Raised **test coverage** from **50%** to **83%** by writing **unit** and **integration tests** for core services and workflows.

Backend Engineer (FastAPI) · Isfahan Moghadam, Iran — 2025 – 2026

- Designed and built a **recruitment and job-matching service** and an **online assessment platform** with **FastAPI** on **Clean**, **Layered** and **Hexagonal Architecture**.
- Focused on reusability, testability, reliability and **idempotency**; the service supports **800+** users.
- Built **JWT** authentication with **access/refresh tokens**, **token rotation**, **token revocation** and **refresh-token reuse detection** to catch replay of revoked tokens.
- Developed business logic and the QC **admin panel** with **Django** and **DRF**; implemented background processing and notifications with **Celery** and **RabbitMQ**, and scheduled jobs (including automated database backups) with **Celery Beat**.

EDUCATION & LEARNING

B.Sc. in Computer Engineering · Shahrekord University

Currently learning: Machine Learning, Deep Learning, Statistical Learning, Model Optimization and Computer Vision.